

Competitive Exam Practice 7 — Answer Key

Microprogrammed Control Unit Design

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- Q1 (D) all of the above.** Horizontal microprogramming dedicates one bit per control signal, so each bit directly *is* the signal (no decoder needed), which is exactly why the microinstruction word is wider than the encoded-field vertical alternative. See the provenance note in the questions file for this answer's verification status.
- Q2 (A).** The 26-bit microinstruction splits as $13 + X + Y$. The MUX has 8 status-bit inputs, so $Y = \lceil \log_2 8 \rceil = 3$. Hence $X = 26 - 13 - 3 = 10$ bits. The next-address field addresses control memory, so the number of words is $2^{10} = 1024$.
- Q3 (C).** With at most two of 25 signals active, encode the (up to) two active signals in two independent fields, each needing $\lceil \log_2 25 \rceil = 5$ bits: $5 + 5 = 10$ bits. (A 2-bit field could not address 25 signals; 25 and 12 are not minimal.)
- Q4 (C).** Field (a) can take one of four states (none or one of 3), needing $\lceil \log_2 4 \rceil = 2$ bits. Field (b) must allow any subset of six micro-operations (each on/off), needing 6 bits. Total $= 2 + 6 = 8$ bits.
- Q5 (C).** Vertical encoding groups mutually-exclusive signals into short encoded fields, which must be expanded by a decoder before the datapath sees valid one-hot signals — that decode step is exactly the extra delay vertical encoding pays for its narrower control word.
- Q6 (B) Mapping unit.** The mapping unit converts the opcode in IR into the starting microaddress of that instruction's microprogram and loads it into MPC; control memory only supplies control words *after* MPC already points at the right row.
- Q7 (B).** Hardwired control has no memory-read delay at all (pure gates); horizontal microprogramming reads a wide control word directly with no decode step; vertical microprogramming pays an additional decoder delay on top of the memory read, making it the slowest of the three.